

Safety and Reliability Training

Module 2: RELIABILITY ENGINEERING

Submodules 2.0 & 2.A

RELIABILITY ENGINEERING INTRODUCTION

Rubén Martínez – Jorge Velasco





Objectives of this training

- ✓ To grasp the importance and potential uses of the Reliability Engineering.
- ✓ To have a practical knowledge of the fundamentals of this area.
- ✓ To understand reliability prediction concept, process and factors.
- ✓ To have an overview of MIL-HDBK-217 guide, methods and ANSI/VITA complement.



Outline

☐ RAM

- Concepts
- Benefits

☐ Reliability

- Features
- Metrics
- System reliability

☐ Maintainability

☐ Availability

☐ Reliability Prediction

☐ Failure Mode and Effect (and Criticality) Analysis (FMEA/FMECA)

☐ Reliability Block Diagrams (RBDs)



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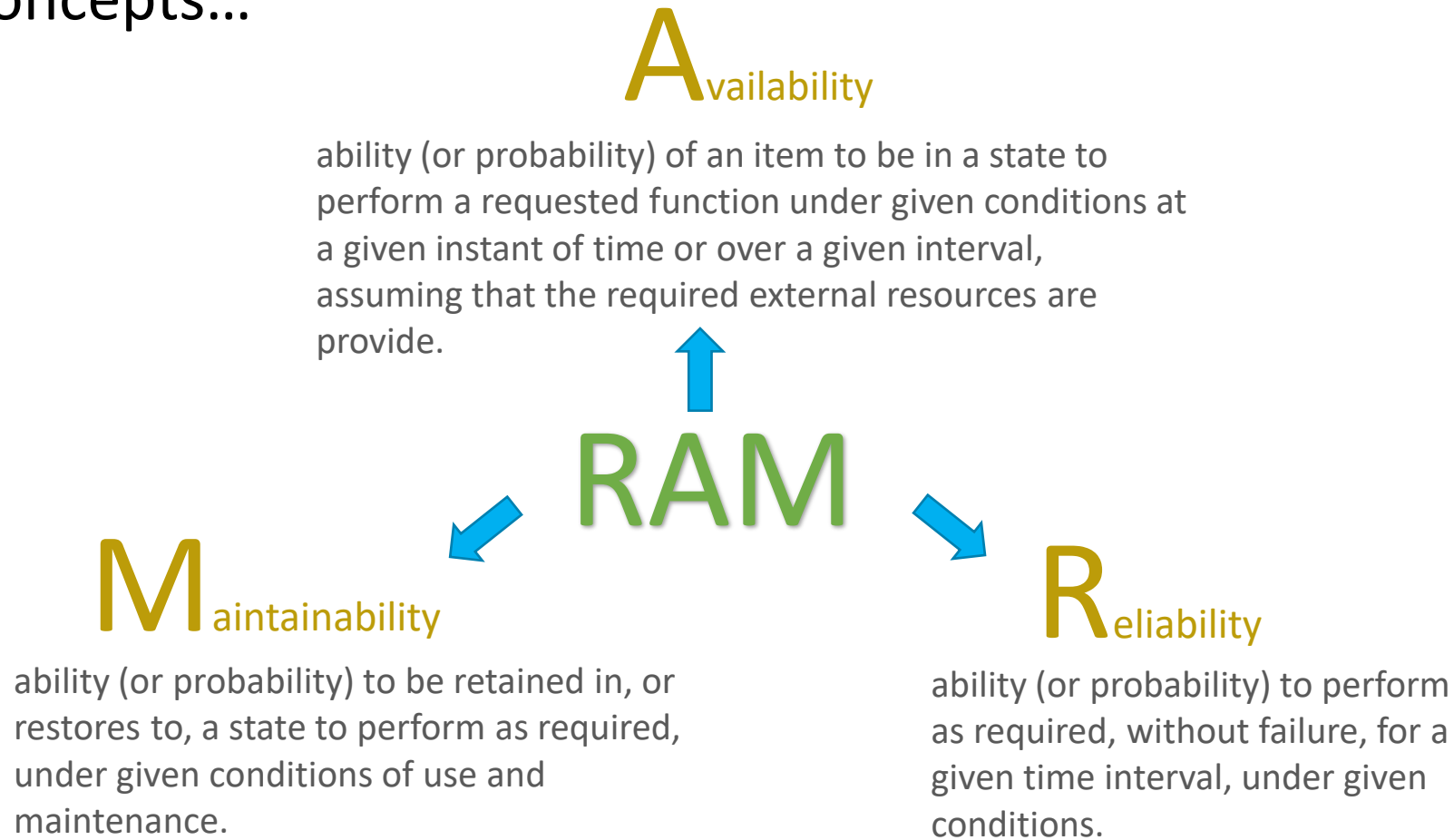
☐ Reliability Prediction

☐ Failure Mode and Effect (and Criticality) Analysis (FMEA/FMECA)

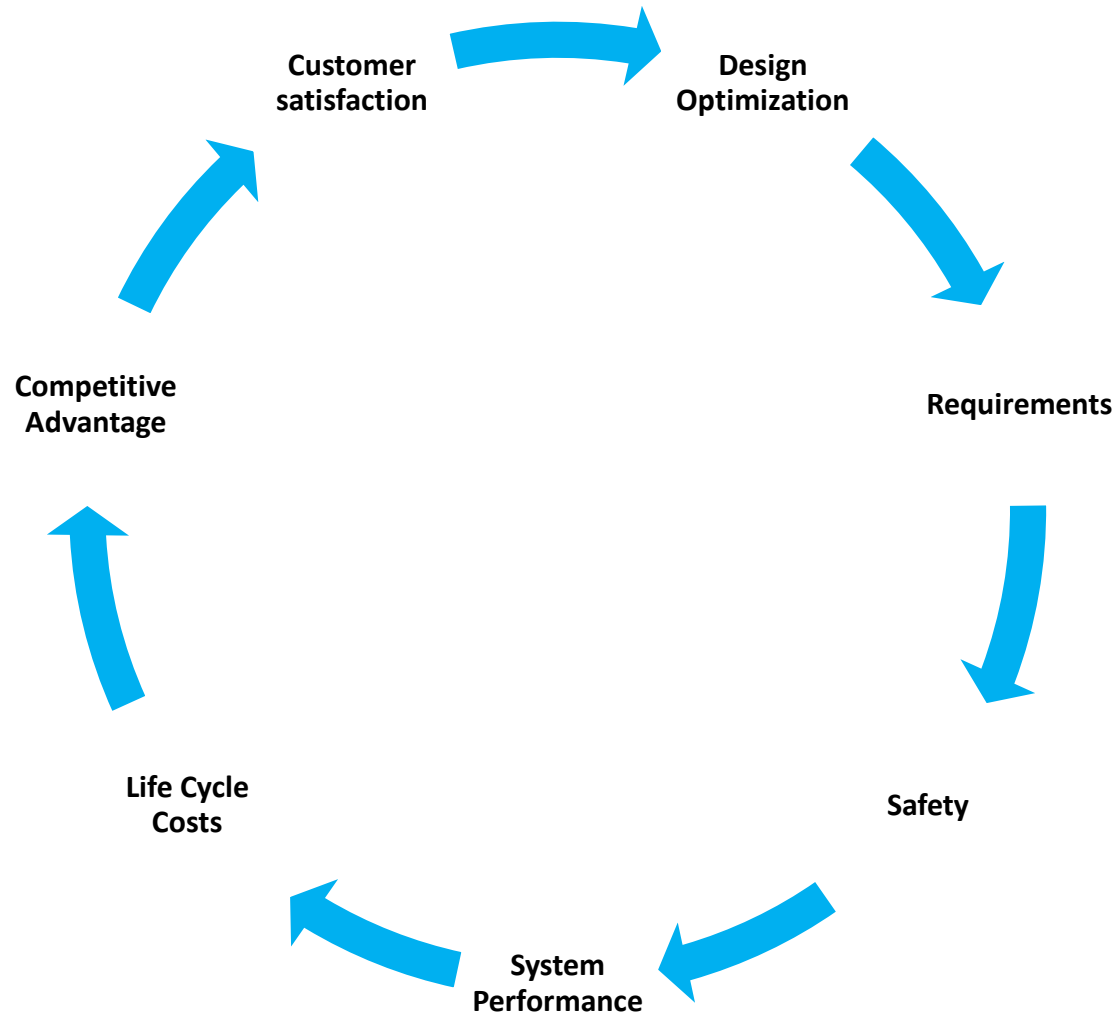
☐ Reliability Block Diagrams (RBDs)

RAM CONCEPTS

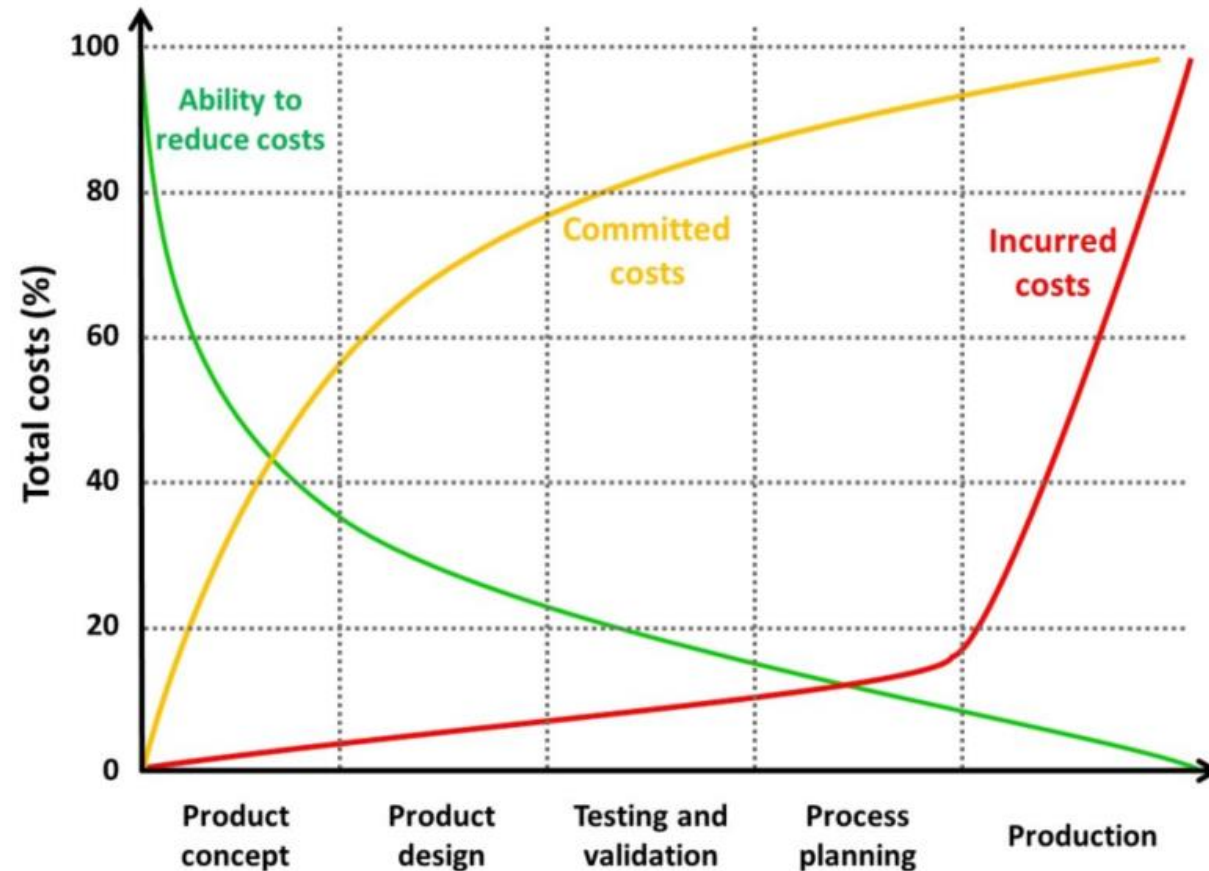
- Some concepts...



RAM Benefits (I)



RAM Benefits (II)



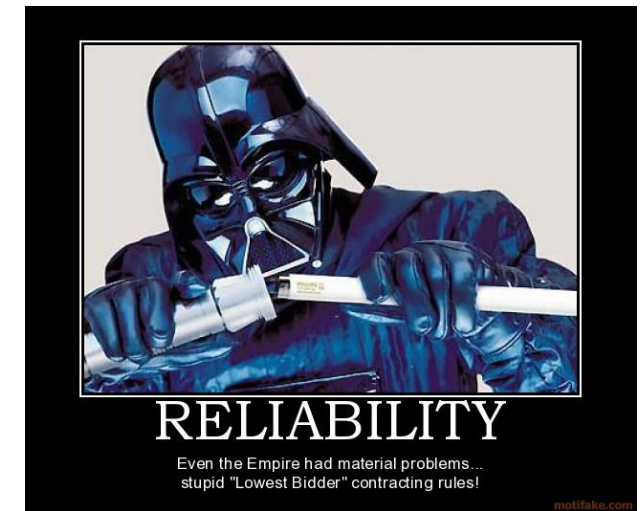


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Reliability

- ❑ **Reliability** can be defined as the probability that a given item will perform its intended function with no failures for a given period of time under a given set of conditions.
- ❑ System or component **reliability** can be predicted at design stage, estimated from test or measured from field data.

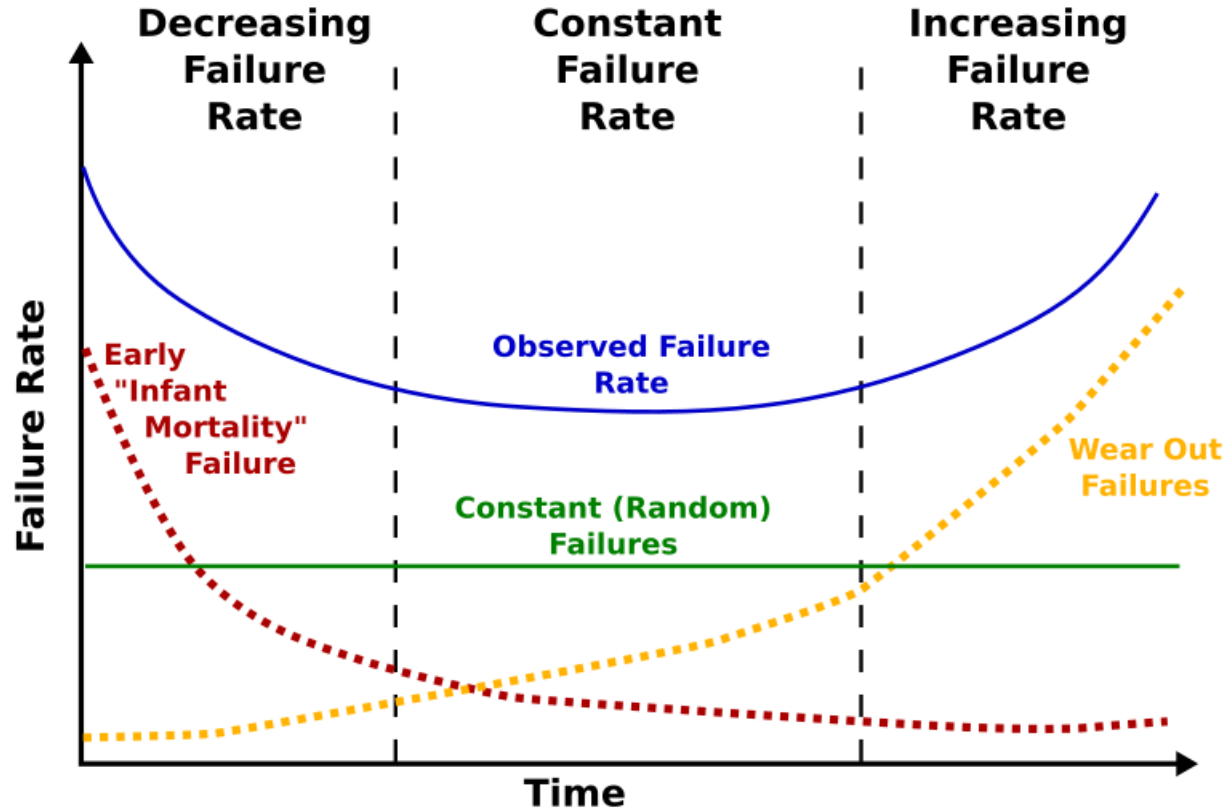


Reliability Features

- ❑ **Probability** -> It is important to note that this is a statistical and expected value, not guaranteed.
- ❑ An **item** can be hardware (system, subsystem, and component), software, and/or human.
- ❑ The **intended function and conditions** (i.e., the operating and environmental conditions) have to be defined by the customer and understood by the reliability engineer.
- ❑ The “given period of time,” commonly referred to as the **mission time**, can be clock hours or cycles.



Main Reliability Metrics (I)



- The first region has a decreasing failure rate due to early failures.
- The middle region is a constant failure rate due to random failures.
- The last region is an increasing failure rate due to wear-out failures.

For electronic components, failure rate is considered to be constant

Main Reliability Metrics (II)

- ❑ **Reliability function (R)** -> probability of survival to time t.

$$R(t) = e^{-\lambda t} \quad (\text{for exponential failure distribution})$$

- ❑ **Failure rate (λ)**

- number of failures per unit time or
- the limit, if this exists, of a ratio of the conditional probability that the instant of time, T, of a failure of an item falls within a given time interval, (t, t + Dt) and the length of this interval, Dt, when Dt tends to zero, given that the item is in an up state at the beginning of the time interval.

- ❑ **Mean time to/between failures (MTTF/MTBF)**

If failure rate (λ) can be considered constant: $MTTF = \frac{1}{\lambda}$

Main Reliability Metrics (III)

- ❑ **Failure function (F)** -> probability of failure at time t

$$F(t) = 1 - R(t) = 1 - e^{-\lambda t} \quad (\text{for exponential failure distribution})$$

- ❑ **Failure rate (λ) as ratio of failure conditional probability**

- Assumption: $t \ll \text{MTBF} = \frac{1}{\lambda} \rightarrow \lambda t \ll 1$

$$F(t) = 1 - R(t) = 1 - e^{-\lambda t} = 1 - \left(1 - \lambda t + \frac{(\lambda t)^2}{2} - \frac{(\lambda t)^3}{6} + \dots\right);$$
$$F(t) \approx \lambda t$$

Since $T_2 = T_1 + \Delta t$ and no failure before $T_1 \rightarrow$

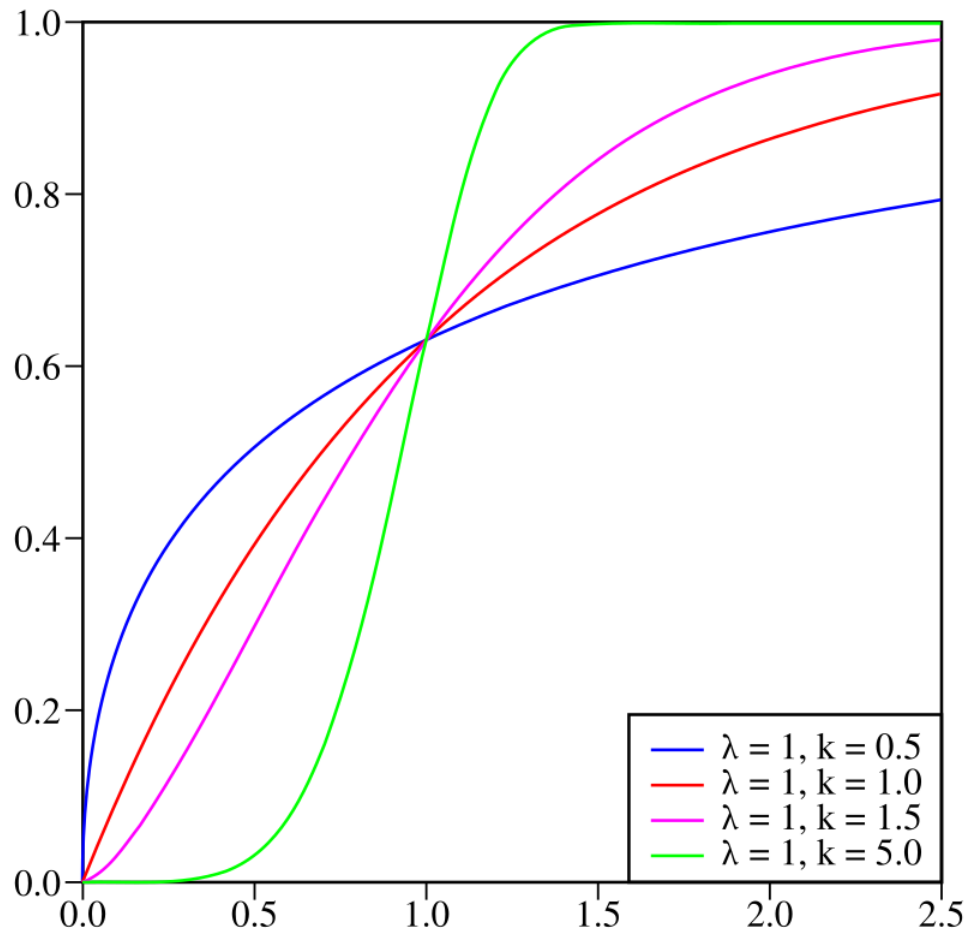
$$\frac{F(\Delta t)}{\Delta t} \approx \lambda$$

- ❑ **A given item will work up to its MTBF with a confidence level of 37%.rate**

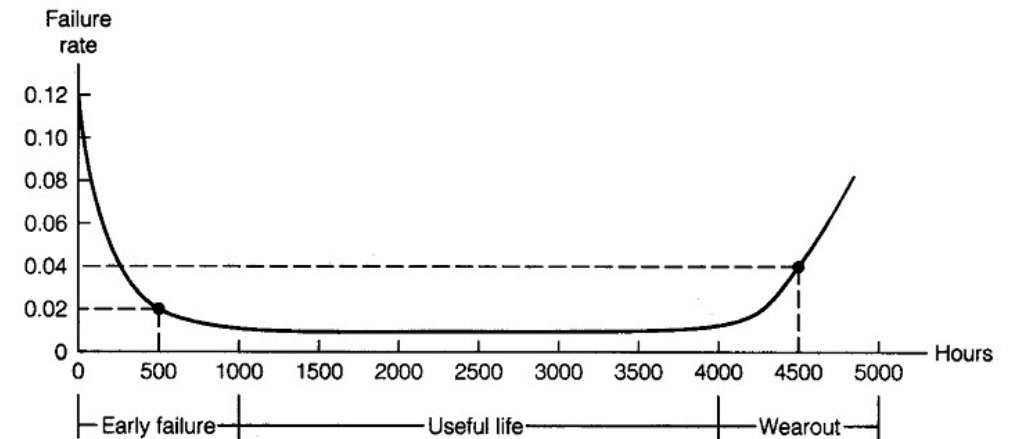
$$R(t) = e^{-\lambda t} = e^{-\frac{t}{\text{MTBF}}} = e^{-1} = 0,37$$

Main Reliability Metrics (IV)

❑ **Weibull distribution** -> probability of failure (F)



Bathtub curve for second Weibull parameter (k) equal to 1 (exponential distribution)



Main Reliability Metrics – Example 1

- An amplifier has an exponential time-to-failure distribution with a failure rate of 8% per 1000 hours. What is the reliability of the amplifier at 5000 hours? Find the mean time to failure.

Solution The constant failure rate λ is obtained as

$$\lambda = 0.08/1000 \text{ hours} = 0.00008/\text{hour}$$

The reliability at 5000 hours is

$$R(t) = e^{-\lambda t} = e^{-(0.00008)(5000)} = e^{-0.4} = 0.6703$$

The mean time to failure is

$$\text{MTTF} = 1/\lambda = 1/0.00008 = 12,500 \text{ hours}$$

Main Reliability Metrics – Example 2

- What is the highest failure rate for a product if it is to have a probability of survival (i.e., successful operation) of 95% at 4000 hours? Assume that the time to failure follows an exponential distribution.

Solution The reliability at 4000 hours is 0.95. if the constant failure rate is given by λ , we have

$$R(t) = e^{-\lambda t} \quad \text{or} \quad 0.95 = e^{-\lambda(4000)}$$

This yields

$$\lambda = 0.0000128/\text{hour} = 12.8/10^6 \text{ hours}$$

Thus, the highest failure rate is $12.8/10^6$ hours for a reliability of 0.95 at 4000 hours.

System Reliability: Components in series

- ❑ In a system with **components in series**, for the system to operate, each component must operate.



- It is assumed that the components operate independent of each other (the failure of one component has no influence on the failure of any other component).
- If there are n components in series, the system reliability is:

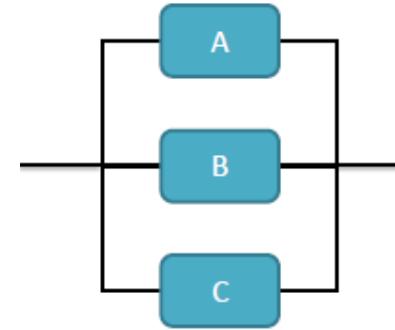
$$R_s = R_1 \times R_2 \times \dots \times R_n$$

- If the system components can be assumed to have a time to failure given by the exponential distribution and each component has a constant failure rate:

$$\begin{aligned} R_s &= e^{-\lambda_1 t} \times e^{-\lambda_2 t} \times \dots \times e^{-\lambda_n t} \\ &= \exp \left[- \left(\sum_{i=1}^n \lambda_i \right) t \right] \end{aligned}$$

System Reliability: Components in parallel

- ❑ In a system with **components in parallel**, the system operates as long as at least one of the parallel components operates.



- It is assumed that the components operate simultaneously, randomly and independently of each other, the probability of failure of each component is given by $F_i = 1 - R_i$.

- The system fails only if all the components fail. Thus, the probability of system failure is:

$$\begin{aligned} F_s &= (1 - R_1)(1 - R_2) \cdots (1 - R_n) \\ &= \prod_{i=1}^n (1 - R_i) \end{aligned}$$

- The reliability of the system is:

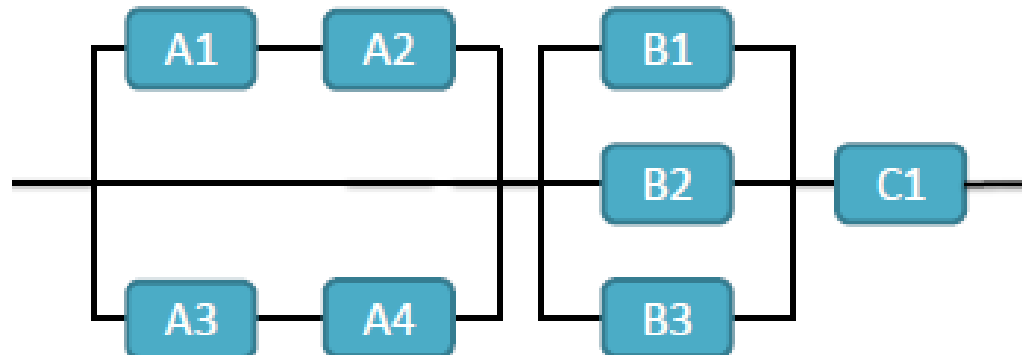
$$\begin{aligned} R_s &= 1 - F_s \\ &= 1 - \prod_{i=1}^n (1 - R_i) \end{aligned}$$

System Reliability

- ❑ The reliability of each component and the configuration (architecture) of the system consisting of these components determines the **system reliability**.
- ❑ One common approach for increasing the reliability of the system is through redundancy in design, which is usually achieved by placing components in parallel: As long as one component operates, the system operates.

Components in series and parallel

- ❑ Complex systems often consist of components that are both in series and in parallel.
- ❑ Reliability calculations are based on the concepts discussed previously, assuming that the components operate independently.





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Maintainability

- ❑ Maintainability is the probability $M(t)$ that a malfunctioning system can be restored to a correct state within time t .
- ❑ It is a measure of the speed of repairing an element or system after the occurrence of a failure.
- ❑ Its main metric is the **Mean Time to Repair (MTTR)**.



Maintenance types

- ❑ **Corrective Maintenance:** Maintenance tasks carried out to correct or mitigate a failure or its effects (corrective or palliative), either in an immediate or in a deferred way.
- ❑ **Preventive Maintenance:** Maintenance tasks carried out before a situation that needs to be avoided occurs. It can be:
 - based on a plan (**scheduled**)
 - based on data (**predictive**)/condition of the equipment (**on condition**).
 - *Real-time monitoring of equipment condition to predict when a failure is likely to occur and take action before it happens. Common technologies: Sensors, vibration analysis, infrared thermography, wear monitoring, etc..



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Availability (I)

- ❑ Essentially, it is a measure of how often the system is **operating** (and available) compared to the total operating time required or expected. (ex. 99.995%).
- ❑ The availability of a system depends not only on how frequently it becomes inoperable (Reliability) but also how quickly it can be repaired (Maintainability).



Availability (II)

Measure	Equation	Description
Inherent Availability	$A_i = \frac{MTBF}{MTBF + MTTR} * 100\%$	- Where MTBF is the mean time between failure and MTTR is the mean time to repair (corrective only) - Reflects the percent of time a product would be available if no delays due to maintenance, supply, etc. (i.e., not design-related) were encountered
Achieved Availability	$A_a = \frac{MTBM}{MTBM + MTTR_{Active}} * 100\%$	- Where MTBM is the mean time between maintenance (preventive and corrective) and MTTR_{Active} is the mean time to accomplish preventive and corrective maintenance tasks - Similar to A_i except that preventive and corrective maintenance are included
Operational Availability	$A_o = \frac{MTBM}{MTBM + MDT} * 100\%$	- Where MTBM is the mean time between maintenance (preventive and corrective) and MDT is the mean downtime, which includes MTTR_{Active} and all other time involved with downtime, such as delays - Similar to inherent availability but includes the effects of maintenance delays and other non-design factors. - A_o reflects the totality of the inherent design of the product, the availability of maintenance personnel and spares, maintenance policy and concepts, and other non-design factors, whereas A_i reflects only the inherent design.
Uptime Ratio	$A = \frac{Uptime}{Uptime + Downtime} * 100\%$	- Uptime is the time that the product is in the customer's possession and works; downtime is the total number of hours that the product is not operable/usable - Uptime Ratio is time-dependent; the time period over which the measurement is made must be known



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Reliability Prediction (I)

- ❑ To obtain an estimation of a significative reliability indicator, i.e. failure rate, MTBF...etc.
- ❑ Reliability of a system or product can be predicted from knowledge of both the reliability of all of its components and their mutual relationships (series, parallel..etc.). It is usually based on established models such as:
 - MIL-HDBK-217F
 - FIDES
 - TELCORDIA



Reliability Prediction (II)

- ❑ Item reliability can also be predicted by life tests by testing a large quantity of it at an increased stress above the nominal one (accelerated life testing).
- ❑ Reliability can be inferred from field data samples subjected to a statistical analysis with a certain level of confidence.
- ❑ Finally, item failure rate can be looked up in specific libraries, such as NPRD, EPRD...etc, according to its classification. However, there is usually a considerable dispersion of data.

Reliability Prediction (III)

Introduction

- ☐ At the early design stage.
- ☐ Verify that the proposed architecture will achieve the imposed reliability objectives (MTBF - Failure rate λ), from a statistical approach.
- ☐ Identify the critical components for reliability (potential design weaknesses).
- ☐ Evaluate the effect of the operating conditions on the failure rate of the components (Environment, Temperature, etc).
- ☐ Evaluate others design options.
- ☐ Perform other RAMS analysis (FME(C)A, FTA, etc).

Reliability Calculation Guides (I)

- ❑ To perform a reliability prediction analysis, a guide is employed
- ❑ Each Reliability Calculation guide offers a set of mathematical formulas to model and calculate the failure rate of a variety of components that make up a product or system.
- ❑ The formulas in the Reliability Calculation guides take into account the factors that affect the reliability of devices*:
 - Operating stresses
 - Temperature
 - Environment
 - Quality level
 - *Others

*The factors used vary depending on the type of component being analysed and the guide used.

Reliability Calculation Guides (II)

Make a list of all the components of your system (BOM)



Find the appropriate equation in the guide for each component



Evaluate the factors needed to compute the failure rate of each component



Perform the failure rate calculation for the system

Reliability Calculation Guides (III)

- ❑ There are several widely accepted Reliability Calculation guides
 - MIL-HDBK-217 (Part Stress, Parts Count)
 - ANSI/VITA 51.1 (a MIL-HDBK-217 Complementary)
 - FIDES
 - NSWC Mechanical
 - Others (Telcordia SR-332, 217PLUS)

- ❑ Additionally, there are component databases like NPRD (Non-electronic Parts Reliability Data).

MIL-HDBK-217 (I)

- In Military & Defense industries, the Mil-HDBK-217 is widely used.
 - Parts Count Method
 - Part Stress Method
 - Reliability formulas for different types of components
- (chapter 5 to chapter 23)

For Stress Method, see the use of Mil-HDBK-217 + ANSI/VITA template on the [Plantilla cálculos Reliability MIL-HDBK-217 | Anzen Wiki \(mooo.com\)](#)

NOT MEASUREMENT SENSITIVE

MIL-HDBK-217F

2 DECEMBER 1991

SUPERSEDING

MIL-HDBK-217E, Notice 1

2 January 1990

MILITARY HANDBOOK

RELIABILITY PREDICTION OF ELECTRONIC EQUIPMENT



MIL-HDBK-217 (II)

- Deals with different types of environments.
- Conservative results (especially pessimistic with non-military components).

Environment	Symbol	Nominal Environmental Conditions
Ground, Benign	G _B	Nonmobile, temperature and humidity controlled environments.
Space, Flight	S _F	Earth orbital. Approaches Ground Benign conditions. Vehicle neither under powered flight nor in atmospheric reentry.
Ground, Fixed	G _F	Conditions less than ideal to include installation in permanent racks with adequate cooling air and possible installation in unheated buildings.
Ground, Mobile	G _M	Conditions more severe mostly for vibration and shock. Equipment installed on wheeled or tracked vehicles.
Naval, Sheltered	N _S	Sheltered or below deck conditions on surface ships and submarines.
Naval, Unsheltered	N _U	Unprotected surface shipborne equipments exposed to weather conditions and salt water.
Airborne, Inhabited, Cargo	A _{IC}	Typical conditions in cargo compartments occupied by aircrew without environmental extremes of pressure, temperature, shock and vibration.
Airborne, Inhabited, Fighter	A _{IF}	Same as A _{IC} but installed on high performance aircraft such as fighters and interceptors.
Airborne, Uninhabited, Cargo	A _{UC}	Uncontrolled areas with environmental extremes of pressure, temperature and shock.
Airborne, Uninhabited, Fighter	A _{UF}	Same as A _{UC} but installed on high performance aircraft such as fighters and interceptors.
Airborne, Rotary Winged	A _{RW}	Equipment installed on helicopters, internally and externally.
Missile, Launch	M _L	Severe conditions of noise, vibration, and other environments related to missile launch, and space vehicle boost into orbit, vehicle re-entry and landing by parachute. Conditions may also apply to installation near main rocket engines during launch operations.
Missile, Flight	M _F	Typical conditions of pressure, vibration and temperature experienced in atmospheric flight to target.

MIL-HDBK-217 - Parts Count (I)

- Early design stage: detailed stress values not available

$$\lambda_{\text{EQUIP}} = \sum_{i=1}^{i=n} N_i (\lambda_g \pi_Q)_i$$

λ_{EQUIP} = Total equipment failure rate (Failures/ 10^6 Hours)

λ_g = Generic failure rate for the i^{th} generic part (Failures/ 10^6 Hours)

π_Q = Quality factor for the i^{th} generic part

N_i = Quantity of i^{th} generic part

n = Number of different generic part categories in the equipment

λ_g : generic failure rate according to a given environment (default temperature) (see tables in appendix A).

π_Q : see ANSI/VITA rules/recommendations

MIL-HDBK-217 - Parts Count (II)

- If the ambient temperature is different from the environment's default temperature, a *correction factor is applied to the generic failure rate.

		Generic Failure Rate - λ_g (Failures/ 10^6 Hours) for Discrete Semiconductors													
Section #	Part Type	Env. T_J (°C) → 50	G_B 60	G_F 65	G_M 60	N_S 65	N_U 75	A_{IC} 75	A_{IF} 90	A_{UC} 90	A_{UF} 75	A_{FW} 50	S_F 65	M_F 75	C_L 60
	DIODES														
6.1	General Purpose Analog	.0036	.028	.049	.043	.10	.092	.21	.20	.44	.17	.0018	.076	.23	1.5
6.1	Switching	.00094	.0075	.013	.011	.027	.024	.054	.054	.12	.045	.00047	.020	.060	.40
6.1	Fast Recovery Pwr. Rectifier	.023	.19	.32	.28	.68	.61	1.4	1.3	2.9	1.1	.012	.50	1.5	10
6.1	Power Rectifier/ Schottky Pwr.	.0028	.022	.039	.034	.082	.073	.16	.16	.35	.13	.0014	.060	.18	1.2
6.1	Transient Suppressor/Varistor	.0029	.023	.040	.035	.084	.075	.17	.17	.36	.14	.0015	.062	.18	1.2
6.1	Voltage Ref/Reg. (Avalanche and Zener)	.0033	.024	.039	.035	.082	.066	.15	.13	.27	.12	.0016	.060	.16	1.3

Temperature Factor - π_T (General Purpose Analog, Switching, Fast Recovery, Power Rectifier, Transient Suppressor)			
T_J (°C)	π_T	T_J (°C)	π_T
25	1.0	105	9.0
30	1.2	110	10
35	1.4	115	11
40	1.6	120	12
45	1.9	125	14
50	2.2	130	15
55	2.6	135	16
60	3.0	140	18
65	3.4	145	20
70	3.9	150	21
75	4.4	155	23
80	5.0	160	25
85	5.7	165	28
90	6.4	170	30
95	7.2	175	32
100	8.0		

$$G_{F@T2} = \frac{\pi_{T2}}{\pi_{T1}} * G_{F@T1}$$

$$G_{F@50} = \frac{\pi_{T50}}{\pi_{T60}} * G_{F@60} = \frac{2,2}{3} * 0,022$$

*ANSI/VITA 51.1 Appendix H.

MIL-HDBK-217 – Stress Part

- The formulas for each chapter (component type) are implemented in the excel template.
- The Voltage Ratio and/or Power Ratio for Resistors, Capacitors and Semiconductors can be calculated from the electronic schematic diagram.
- If unable to calculate, see ANSI/VITA rules and permissions for correction of quality factors (π_Q), Voltage Ratio/Power Ratio, and other considerations.

π -Factors Resistors Stress Part:

$$\lambda_p = \lambda_b \pi_T \pi_P \pi_S \pi_Q \pi_E \text{ Failures}/10^6 \text{ Hours}$$

ANSI/VITA 51.1 (I)

- Its rules, recommendations, and suggestions take into account changes in device technologies.
- More accurately models the “Quality (π Q)” of commercial components from “known pedigree manufacturer”.



Approved ANSI Standard

ANSI/VITA 51.1 – 2013 (R2018)

Reliability Prediction MIL-HDBK-217 Subsidiary Specification

ANSI/VITA 51.1 (II)

- For both Parts Count and Stress Method:
- Chapter 5.0 Integrated circuits: adjust manufacturer's failure rate to operating conditions (environment and temperature) according to permission 2.3.4-1:

$$\lambda_P = \left(1 + \frac{\pi_T - \pi_{TMFR}}{\pi_{TMFR}} + \frac{\pi_E - \pi_{EMFR}}{\pi_{EMFR}} \right) \lambda_{BMFR}$$

- Implemented in the template.

Reliability Prediction - example

The screenshot displays the ANZEN software interface. On the left, a 'System Tree' panel lists components: Industrial Tablet PC, Battery, Motherboard, Microprocessor, Static RAM, Clock Generator, Video Processor, Touchpanel, Memory Board, and Hard Disk Assembly. The main window shows a table with columns: Name, Part Number, System Tree Identifier, Reference Designator, Description, Manufacturer, Failure Rate, Predicted, and MTBF, Predicted. A 'View Calculation Results' dialog box is open, showing 'Prediction Results' for the selected component. The dialog contains a table with 'Value' and 'Result' columns, listing various reliability metrics and their predicted values. At the bottom of the dialog are buttons for 'Excel', 'Print...', 'Close', and 'Help'.

Name	Part Number	System Tree Identifier	Reference Designator	Description	Manufacturer	Failure Rate, Predicted	MTBF, Predicted
Industrial Tablet PC	PC070101	S1	S1	Tuf-Tablet	Tuf-Tablet International	23.200524	43102
Battery	BAT56A04	A5	A5	Li-Ion tablet ...	Superior Power Devices	2.242700	445891
Motherboard	MB060415	A1	A1	Motherboard	iTronics	5.323376	187851
Microprocessor	MIC870A					4.670594	214106
Static RAM	SRA					0.512888	2e+006
Clock Generator	CLK					0.104257	1e+007
Video Processor	VPB					0.035637	3e+007
Touchpanel	TPS					6.550000	152672
Memory Board	MEB					1.355120	737942
Hard Disk Assembly	HDA					7.729327	129377

Value	Result
Failure Rate, Predicted	23.200524
MTBF, Predicted	43102
Reliability, Predicted	0.498568
Availability	0.999997
MTTR	0.127268
Failure Rate, Mission	27.646059
MTBF, Mission	36172
Reliability, Mission	0.436319
Availability, Mission	0.999996

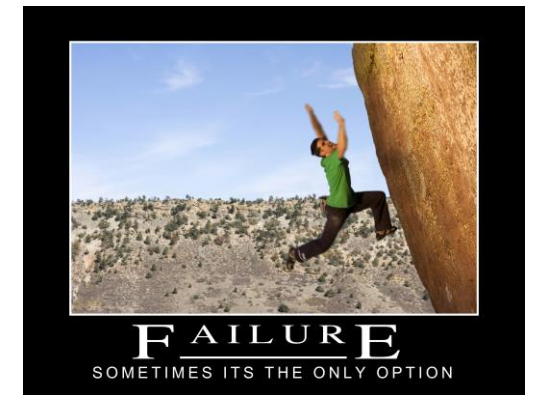


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Failure Mode and Effect (and Criticality) Analysis (FMEA/FMECA)

- ❑ Systematic analysis of postulated component failures, identification of their causes and evaluation of the resultant effects.
- ❑ Because of its thoroughness , it can provide very relevant outcomes but at the same time be very effort-consuming.
- ❑ It can be applied to Systems, products and processes.



Failure Mode and Effect (and Criticality) Analysis (FMEA/FMECA) Benefits

- ☐ Allows to identify and evaluate potential simple failures that may impact System performance, safety or interfaces.
- ☐ Evaluates the capacity to detect failures.
- ☐ Help to optimize maintenance activities.
- ☐ Provides a valuable feedback to design activities and processes.

FMEA/FMECA Datasheets examples

FMEA

ID	Function or Component	Failure mode	Cause	Mission phase	Local Effects	End Effects	Severity	Observable symptoms	Compensating provisions	Recommendations

FMECA

ID	Function or Component	Failure mode	Cause	Mission phase	Local Effects	End Effects	Severity	Probability	Criticality	Observable symptoms	Compensating provisions	Recommendations

Failure Mode and Effect (and Criticality) Analysis (FMEA/FMECA) example

PTC Windchill Quality Solutions - [My Training Project, System: Tablet PC]

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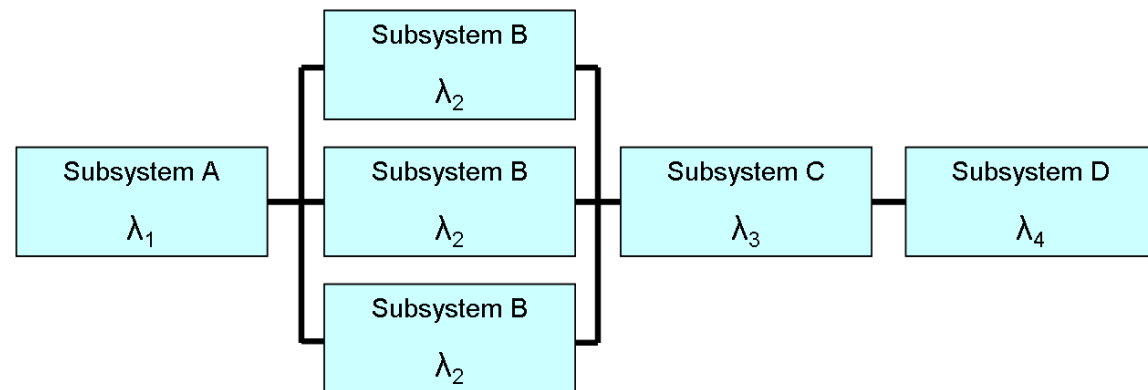


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Reliability Block Diagrams (RBDs)

- ❑ Reliability Block Diagram (RBD) analysis is a RAM methodology for assessing and calculating the reliability, maintainability and availability metrics of complex systems using a graphical depiction of the system.
 - RBDs can be used to optimize parameters, spares, costs, etc.
 - RBDs are created in a visual format, using blocks to represent components or elements of the system and connections between the blocks to define their relationships.



Any questions?



Safety and Reliability
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